

Building a Personal Knowledge Website

Agentic Coding for Geospatial

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Module Overview

This module is not directly related to geospatial analysis, but you will learn useful skills that will help you in your knowledge work.

- Learn how to use AI Agents to organize and manage a corpus of documents.
- You'll learn how to build and manage modern websites using AI Agents.



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Ujaval's Notes and Bookmarks



This site is a collection of my personal notes and bookmarks collected over the years. It is organized to help search and recall specific items or find notes related to a certain topic.

These notes were scattered across many places and hard to manage. Finally, Claude Code helped organize this and build this site. These are now auto-updated weekly from my notes. See the [CLAUDE.md](#) file with processing instructions.

Latest Finds

- [Bhoonidhi MCP Server](#): MCP server that lets AI agents search and download ISRO's Bhoonidhi satellite imagery archive using natural language. [Keywords: MCP Claude Code ISRO India satellite imagery remote sensing] — [More in Claude Code](#)
- [isobands: Xarray to GeoPandas Contours](#): Python library converting 2D xarray DataArrays into filled contour polygons in GeoPandas, spun off from the Reuters Climate Monitor. [Keywords: XArray isobands contours GeoPandas GDAL Reuters] — [More in XArray](#)
- [India Data Atlas](#): Collection of data sources for India with learning resources. [Keywords: data India geospatial socio-economic open data learning resources] — [More in Data](#)

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Latest Finds

Topics

Geospatial Tools

Data Science

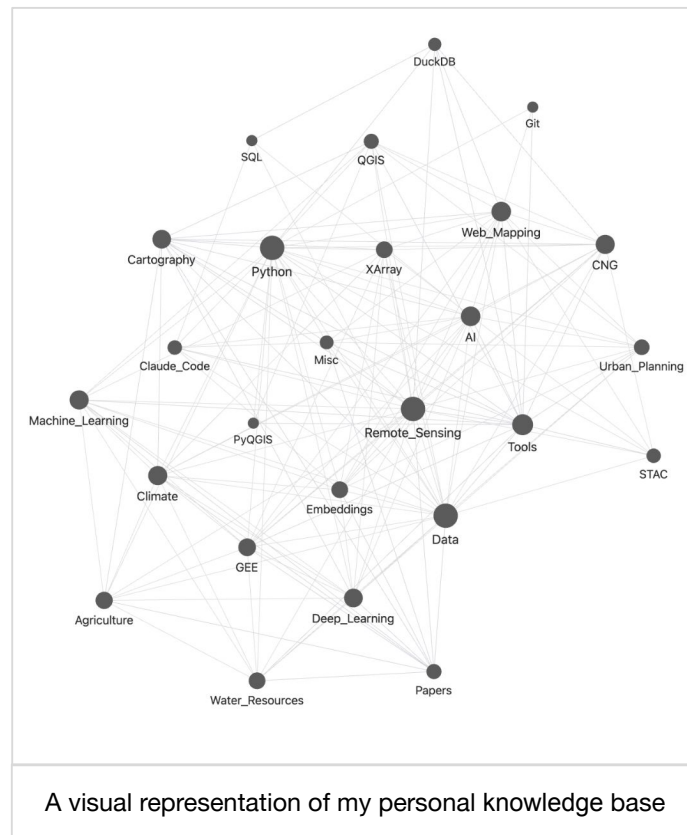
AI & Machine Learning

Domain Applications

Other

What is a Personal Knowledge Base?

- A persistent digital memory system that stores your notes and ideas.
- LLM-powered agents can help organizing and maintaining information,
 - Bring together Information scattered across different places
 - Automatically categorize and curate notes
 - Interlink and augment with missing information
- A ***Personal Knowledge Base*** can help you remember and recall specific information



RAG (Retrieval-Augmented Generation)

- RAG (Retrieval-Augmented Generation) is a framework for creating a searchable database of documents.
- Used by organizations for building chatbots from internal documents.
- How does a RAG work?
 - Read documents and split into chunks.
 - Use an embedding model to generate a unique embedding for that chunk
 - Store the embedding in a vector database.
 - When a user searches for a text, match its embedding from the database
 - Retrieve matching documents.
- RAGs can enable queries like
 - *List all the projects we carried out for client X.*
 - *What are some of the problems we encountered when working in region Y?*

The LLM-Wiki System

- Andrej Karpathy proposed a version of RAG for a personal knowledge base called LLM-Wiki.
- A system that creates an interlinked collection of Markdown documents maintain by an AI-agents.
- The idea resonated with many people
 - **Simple:** Notes stored as plain text/Markdown files. No database required
 - **Easy to maintain:** LLMs help maintain it. Requires little effort from humans.
 - **Easy to search:** LLMs can search, summarize and answer questions.
 - **Evolves over time:** The corpus is constantly updated with new information.
 - **Free and open:** No proprietary components or lock-in.

Who should use LLM-Wiki?

- ✓ If you take lots of notes
- ✓ If you keep a journal with your ideas/learnings/goals and want a better system to organize them.
- ✓ If you read a lot of research papers and books and want to help synthesize the information better
- ✓ If you already use a note-taking system and want to move to a modern AI-enabled system



[Karpathy's LLM Wiki](#)
[Full Beginner Setup Guide](#)

A Simplified Personal Knowledge Base

- We will implement a simplified version of the LLM-Wiki focused on ***organizing your bookmarks and notes***.
- Focus on helping you remember and recall information.
- Setup a note taking system using Obsidian and plain Markdown files.
- Claude Code reads the notes and organizes them.
 - Adds description, tags, topics
- Your personal knowledge base is a collection of interlinked notes.
 - Visualize, search and retrieve specific items
- Publish the knowledge base as a searchable mobile-friendly website.
- Claude Code updates the archive and the website automatically as you add new notes.

Personal Knowledge Base vs. LLM-Wiki

	Simplified Personal Knowledge Base	LLM-Wiki
Focus	Remember and recall specific items	An AI Second Brain that you can offload some of your cognitive load
Inputs	URLs and Notes	Documents, Paper, Meeting Notes, ...
Tasks	Organizing	Organizing, synthesizing, cross-referencing, updating
Ask questions	No	Yes
Suggest ideas	No	Yes
Updates	Add new information only	Identifies contradictions, missing links and seeks new information



Publishing Your Notes Website

How to build a website?

- Websites are built using three core languages
 - **HTML**: Used to define the structure of each page and its content
 - **CSS**: Used to design layout, colors, fonts, responsiveness etc.
 - **Javascript**: Used to make pages interactive
- There are two main types of websites
 - Dynamic Websites
 - Static Websites

Dynamic Websites

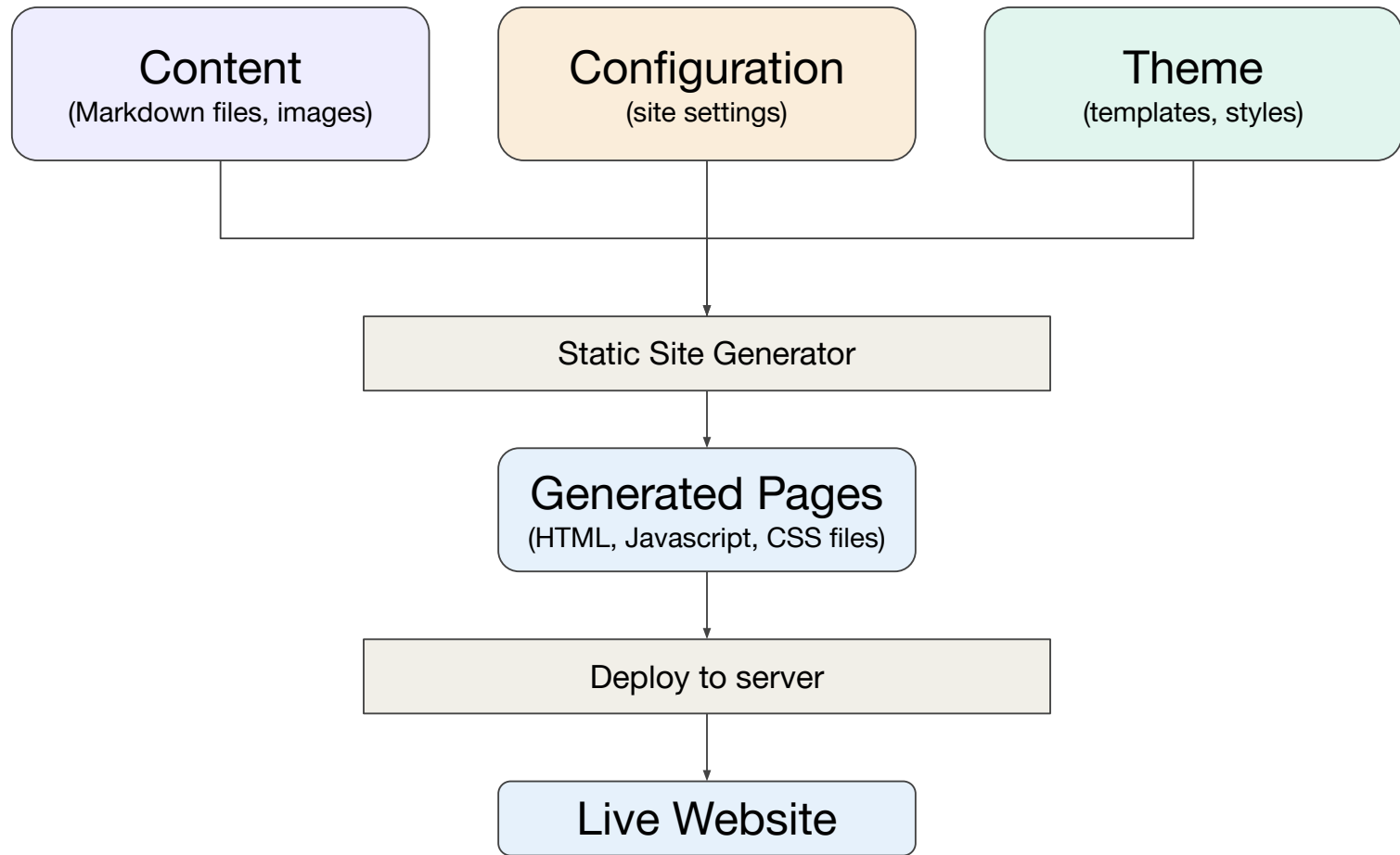
- Dynamic websites store data in a database and have a server that generates pages on-the-fly based on user request.
- When a visitor visits a page, the server queries the database, and assembles the page using HTML, CSS and Javascript.
- Dynamic websites are ideal for
 - Websites that store and process user data. i.e. Banking, E-Commerce, Apps
 - Websites that needs to serve large datasets and need to pre-process it. i.e. Dashboards, Open Data Portals.
 - Websites that needs to serve real-time or near real-time data. i.e. News, Weather
- Dynamic sites need a backend (database, server..) and a frontend (HTML, CSS, Javascript)

Static Sites

- A static site is a collection of pre-built HTML, CSS, and JS files
- Same set of pages are shown to all users - no server side processing is done.
- Static sites are ideal for hosting content that does not require a login
 - i.e. Portfolios, Blogs, Documentation
- Hosting static sites is very cheap and many providers give free hosting.
 - I.e. GitHub Pages

How to build a static website?

- Most static sites are created using **Static Site Generators**
- Uses separation of content and styling.
 - You write your content in content files (in simple markup languages like Markdown)
 - You configure settings in a central configuration file
 - You use a theme (with its CSS and Javascript files) that defines how the pages look.
- The content generator program builds your website
 - Creates fully functional website pages from your content, configuration and theme.
 - When you make a change to your content, you run the generator and a new set of pages are generated.
- The generated pages are hosted on a static server which serves them as a website.



Building your notes website

- We will use [MkDocs](#) static site generator.
- Use your Obsidian notes vault as the content for the website.
- Use the [personal-notes-website-template](#) which has the necessary configuration to turn the Obsidian vault into a MkDocs website.
- Use Claude Code to create a new GitHub repository and publish the website.
- GitHub provides free hosting via GitHub Pages and free deployment via GitHub Actions.
- Claude Code manages the entire process of creation and maintenance.
 - Add a note, update Obsidian vault, push changes to GitHub, deploy to GitHub Pages.